1. Explain “Integral”.

2. Express the limit as a definite integral on the given interval.

$$\lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{2x_i^* + (x_i^*)^2}\, \Delta x, \quad [1, 8]$$

3. Prove the Mean Value Theorem for Integrals by applying the Mean Value Theorem for derivatives to the function

$$F(x) = \int_a^x f(t)\, dt.$$

THE MEAN VALUE THEOREM Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.

2. f is differentiable on the open interval (a, b).

Then there is a number c in (a, b) such that

$$\boxed{1} \qquad f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$\boxed{2} \qquad f(b) - f(a) = f'(c)(b - a)$$

4. We recommend you to have the class survey via TWINS. Please, visit TWINS and click “Enquete”. And then find for CALCULUS I (2023FallA). Always welcome your comment.